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SIEM SECURITY ANALYSIS REPORT

Generated: 2026-03-31 23:27:00 | Log Source Type: Postfix

Total Events	1000
Parsed Successfully	1000 (100%)
Risk Level	MEDIUM
ML Detected Threats	0

Overall Risk Score: 40/100 (MEDIUM)

1. EXECUTIVE SUMMARY

This comprehensive security report presents the analysis of **1,000** log events from the SIEM system. The analysis successfully parsed **1,000** events (100% success rate) with **0** events failing to parse.

■ **No ML-detected threats** were found in the analyzed log sample. The security posture appears normal.

2. AI-POWERED THREAT ANALYSIS

This section contains detailed threat analysis generated by our AI SOC Analyst.

SIEM Security Analysis Report

Date: March 19, 1900 (Based on log data)

Analyst: AI Security Analyst

1. EXECUTIVE SUMMARY:

The organization's email infrastructure is currently under potential attack, exhibiting signs of brute-force attempts, threat intelligence indicators, and potential spam activity. While the overall log volume is dominated by informational messages, the presence of alerts necessitates immediate investigation and remediation to prevent further compromise.

2. KEY FINDINGS:

| Finding | Severity | Description |

|---|---|---|

| Multiple Brute-Force Attempts | HIGH | Detected brute-force attacks against the mail server, potentially targeting user credentials. |

| Threat Intelligence Match | HIGH | Detected traffic or activity matching known threat intelligence indicators, suggesting potential compromise or communication with malicious actors. |

| Potential Spam Activity | MEDIUM | Identified potential spam activity originating from or traversing the mail server, which could lead to reputational damage and blacklisting. |

| High Connection Rate from Single IP | MEDIUM | IP address 45.115.119.123 exhibited a high connection rate (11/3600s), potentially indicating automated scanning or abusive behavior. |

| Internal Email Sending Large Attachments | LOW | Emails originating from finance@tmv.edu.in are sending attachments of ~150KB, which could be normal, but warrants investigation as a potential exfiltration attempt or vector for malware delivery. |

3. ATTACK CHAIN ANALYSIS:

Based on the alerts and log data, a potential attack chain can be inferred:

* **Initial Access:** The attacker(s) are attempting to gain initial access through brute-force attacks against the mail server (Postfix). This aligns with **MITRE ATT&CK; T1110 - Brute Force**.

* **Persistence (Potential):** If the brute-force attacks are successful, the attacker could establish persistence by creating new accounts or modifying existing ones.

* **Command and Control (Potential):** The threat intelligence match suggests potential communication with a command and control (C2) server, which could be used to control compromised systems and exfiltrate data. This aligns with **MITRE ATT&CK; TA0011 - Command and Control**.

* **Exfiltration (Potential):** The spam activity could be a precursor to data exfiltration or a means to distribute malicious payloads. The large attachments being sent by the finance account could also be a form of exfiltration.

4. THREAT INTELLIGENCE:

* **Suspicious IPs:**

* `209.85.210.52`, `209.85.161.69`: These IPs are associated with Google. While not inherently malicious, their presence in the top sources alongside other suspicious IPs warrants further investigation. They may be used for relaying spam or phishing emails.

* `45.115.119.123`: High connection rate suggests potential scanning or abusive behavior. This IP should be investigated and potentially blocked.

* **Attacker TTPs (Tactics, Techniques, Procedures):**

* **Brute-Force Attacks:** Attempting to guess user credentials to gain unauthorized access.

* **Spam Distribution:** Utilizing the mail server to send unsolicited emails, potentially containing malicious links or attachments.

* **Potential Command and Control Communication:** Communicating with external servers for command and control purposes (based on threat intelligence alert).

5. RISK ASSESSMENT:

Current Risk Level: HIGH

* **Justification:** The combination of brute-force attacks, threat intelligence matches, and potential spam activity indicates a significant risk to the organization's email infrastructure and overall security posture. Successful brute-force attacks could lead to account compromise, data breaches, and reputational damage. The threat intelligence match suggests a potential connection to known malicious actors, further increasing the risk.

****6. RECOMMENDATIONS:****

1. ****Implement Multi-Factor Authentication (MFA):**** Enforce MFA for all email accounts to mitigate the risk of successful brute-force attacks. This is a critical step to protect against compromised credentials.
2. ****Block Suspicious IPs:**** Immediately block IP address `45.115.119.123` at the firewall level to prevent further abusive behavior.

3. **Investigate Threat Intelligence Match: Conduct a thorough investigation of the threat intelligence alert to identify the specific threat actor and their potential impact on the organization.**

4. ****Strengthen Password Policies:**** Enforce strong password policies, including minimum length, complexity requirements, and regular password changes.
5. ****Implement Rate Limiting:**** Configure rate limiting on the mail server to prevent brute-force attacks and spam activity.
6. ****Review Email Filtering and Anti-Spam Configuration:**** Ensure that email filtering and anti-spam solutions are properly configured and up-to-date to detect and block malicious emails.
7. ****Investigate Finance Account Activity:**** Investigate the email activity of the `finance@tmv.edu.in` account to determine the legitimacy of the large attachments being sent.

****7. INCIDENT RESPONSE:****

1. **Containment:**

- * Isolate any potentially compromised systems.
- * Block suspicious IP addresses at the firewall.
- * Disable compromised accounts.

2. **Eradication:**

- * Remove any malware or malicious files from compromised systems.
- * Reset passwords for all affected accounts.
- * Review and update security configurations.

3. **Recovery:**

- * Restore systems from backups if necessary.
- * Monitor systems for any signs of further compromise.

*** Implement the recommendations outlined in this report.**

4. ****Post-Incident Activity:****

- * Conduct a thorough review of the incident to identify any weaknesses in the organization's security posture.
- * Update security policies and procedures as needed.
- * Provide security awareness training to employees.

This report provides an initial assessment of the security situation based on the provided log data. Further investigation and analysis may be required to fully understand the scope and impact of the detected threats.

3. THREAT INTELLIGENCE & IOCs

Top Source IP Addresses (by event count)

Rank	IP Address	Events	Risk Assessment
1	127.0.0.1	245	HIGH
2	209.85.210.52	16	LOW
3	209.85.161.69	16	LOW
4	45.115.119.123	11	LOW
5	76.223.181.129	11	LOW
6	45.115.119.124	9	LOW
7	40.107.51.105	9	LOW
8	185.93.89.95	9	LOW
9	192.168.1.254	8	LOW
10	209.85.218.53	8	LOW

Most Targeted User Accounts

Rank	Username	Events	Priority
1	iqac@tmv.edu.in	18	MEDIUM
2	infocenter@tmv.edu.in	14	MEDIUM
3	finance@tmv.edu.in	11	MEDIUM
4	0.001	10	MEDIUM
5	examination@tmv.edu.in	10	MEDIUM
6	kulguru@tmv.edu.in	8	MEDIUM
7	law@tmv.edu.in	8	MEDIUM
8	administration@tmv.edu.in	8	MEDIUM
9	phd@tmv.edu.in	6	MEDIUM
10	tmvcompscience@gmail.com	6	MEDIUM

5. SECURITY ALERTS

Total alerts generated: 4. Showing top 15 alerts.

Alert #1: bruteforce

Timestamp: 1900-03-18T16:11:13 | **Source:** N/A

Message: Attack detected: bruteforce (confidence: 85%)

Alert #2: threat_intel

Timestamp: 1900-03-18T16:28:39 | **Source:** N/A

Message: Attack detected: threat_intel (confidence: 80%)

Alert #3: spam_activity

Timestamp: 1900-03-18T16:29:07 | **Source:** N/A

Message: Attack detected: spam_activity (confidence: 60%)

Alert #4: bruteforce

Timestamp: 1900-03-18T16:34:43 | **Source:** N/A

Message: Attack detected: bruteforce (confidence: 85%)

6. LOG ANALYTICS & STATISTICS

6.1 Severity Distribution

Severity Level	Event Count	Percentage
CRITICAL	0	0.0%
ERROR	6	0.6%
WARNING	7	0.7%
INFO	987	98.7%
DEBUG	0	0.0%

6.2 Log Type Distribution

Log Type	Count
postfix	1000

6.3 Top Source IP Addresses

Rank	IP Address	Event Count	Activity Level
1	127.0.0.1	245	HIGH
2	209.85.210.52	16	LOW
3	209.85.161.69	16	LOW
4	45.115.119.123	11	LOW
5	76.223.181.129	11	LOW
6	45.115.119.124	9	LOW
7	40.107.51.105	9	LOW
8	185.93.89.95	9	LOW
9	192.168.1.254	8	LOW
10	209.85.218.53	8	LOW

7. RECOMMENDATIONS & REMEDIATION

◆ Critical

Immediately investigate all critical and high-severity alerts. Isolate affected systems if compromise is suspected.

◆ High Priority

Review and block suspicious IP addresses with high event counts. Implement additional monitoring on targeted accounts.

◆ Medium Priority

Enable enhanced logging for critical systems. Review access controls for accounts with failed authentication attempts.

◆ Best Practices

Ensure all systems are patched and up-to-date. Implement MFA for all privileged accounts. Regular review of security logs.

◆ Monitoring

Set up real-time alerts for threshold-based detections. Create correlation rules for multi-stage attack patterns.

◆ Response

Document and test incident response procedures. Ensure backup and recovery mechanisms are functional.

8. APPENDIX

Report Metadata

Report Generated	2026-03-31 23:27:00
Log Source Type	Postfix
Total Lines	0
Parsed Lines	0
Failed Lines	0
ML Attacks Detected	0
Total Alerts	4
Analysis Engine	Cyber Chakshu SIEM v2.0 + AI SOC Analyst

Disclaimer: This report is generated automatically by Cyber Chakshu SIEM system. The analysis is based on log data and machine learning models. Human review is recommended for critical security decisions.